Local Grammars for the Description of Multi–Word Lexemes and their Automatic Recognition in Texts

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Abstract

Most multi–word lexemes (MWLs) allow certain types of variation. This has to be taken into account for their description to be able to recognize them in texts. We suggest to describe their syntactic restrictions and their idiosyncratic peculiarities with local grammar rules, which at the same time permit to express regularities valid for a whole class of MWLs such as word order variation in German. The local grammars can be written in a very convenient and compact way as regular expressions in the formalism IDAREX which uses a two-level morphology. IDAREX allows to define various types of variables, and to mix canonical and inflected word forms in the regular expressions. The finite–state based dictionary look–up system LOCOLEX/COMPASS uses such local grammars to recognize MWLs in English, German and French on–line texts.

1 Introduction

Most texts are rich in multi-word expressions that cannot be properly understood, let alone be processed in an NLP system if they are not recognized as complex lexical units. We call such expressions *multi-word lexemes (MWL)*. These range from idioms (*to rack one's brains over sth*), over phrasal verbs (*to come up with*) and separated particle verbs (in German or Dutch), lexical and grammatical collocations (*to make love, with regard to*, resp.) to compounds (*on–line dictionary*).

Certain MWLs always occur in exactly the same form, like *out of the blue* or *um Haaresbreite* ('almost', lit. for hair's breadth); these can be easily recognised with simple pattern matching techniques. However, it is well known (see for instance, Gross (1982), Brundage et al. (1992), Nunberg et al. (1994)) that most MWLs cannot be treated in the same way as completely fixed patterns, since they may undergo some variation. Only a *subset* of the variations allowed by general syntax rules is valid though: outside that subset, the expression loses its special, idiomatic meaning, either reverting to its literal meaning or losing any significance altogether. Thus it is important that a representation of MWLs licenses only the allowed variations and excludes variations that would lead to the literal reading of the word sequence.

In certain cases, MWLs can even contradict general syntactic rules, as with *by and large* or G:*von Haus aus* ('originally', lit. from house out), where general rules would require an article between the preposition and the count noun. These latter MWLs have to be recognized before parsing because otherwise the parser would fail.

The identification of MWLs is essential for any natural language processing based on lexical information, ranging from intelligent dictionary look–up over concordancing or indexing to machine translation. Therefore, the restricted lexical and syntactic variability of MWLs and their idiosyncratic peculiarities need to be expressed in the computational lexicon in order to be able to recognize the full range of their occurrences. We propose to use local grammars for this, written as a special type of regular expressions (REs) in the finite–state formalism IDAREX which makes use of a two–level morphological lexicon.

2 Variability of MWLs

Basically, we identify four types of variability (see also Fleischer (1982), Brundage et al. (1992), Van der Linden (1993) and Engelke (1994)) that a description of MWLs, both for NLP and for human use, should cover:

- Morphological variation:
  particular words in the MWL undergo certain inflections.
- Lexical variation:
  one or more words can be substituted by other terms without changing the overall meaning of the MWL. These may be near–synonyms but need not be so.
- Modification:
  one of the MWL's constituents can be modified, preserving the idiomatic meaning.
- Structural flexibility:
  this includes phenomena like passivization, topicalization, scrambling, raising constructions etc.

In the following paragraphs we give English, French and German examples to illustrate the various types of variability. Though we do not always give contrastive examples of MWLs for which the same type of variation is *not* possible, the reader should keep in mind that the illustrated variations only apply on a case by case basis to certain MWLs within one language.

Morphological Variation For instance, in *to kick the bucket*, *kick* can be inflected to any tense, number, person, but the noun may only be used in the singular form to preserve the idiomatic meaning. In F:*faire le fou* ('to play the fool', lit. make the fool) the determiner and the noun can appear in their feminine or plural form, i.e. *faire la folle/les folles.* In G:*durchschlagender Erfolg* ('sweeping success', lit. making its mark success), noun and adjective can vary in case and in number, and comparative and superlative form are possible for the adjective, whereas G:*grüne Welle* ('phased traffic lights', lit. green wave) may only vary in case, but neither in number nor in adjective comparison without loosing its idiomatic meaning.

Lexical Variation For instance, in *to sweep sth under the carpet*, the nouns *rug* or *mat* can be substituted for *carpet*, but not *shag rug* or *stair–carpet.* In F:*perdre la tête* ('to lose one's mind', lit. lose the head), the noun can be substituted by *la boule* (lit. ball, coll. head) or *les*

pédales (lit. pedals) without loosing its idiomatic meaning, but not by *la tronche* (lit. mug, coll. head). In G:*sich aufs Ohr legen* ('to hit the hay', lit. onself onto the ear lay), the exchangeable components have no meaning similarities at all: *hauen* (lit. to hit) can be used instead of *legen* (lit. to put/lay down), and *aufs Ohr* (lit. onto the ear) can be replaced by *in die Falle* (lit. into the trap).

Modification For instance, in F:*avoir un coup dans l'aile* ('to be the worse for drink', lit. have a hit in the wing) an adjective can be inserted, and the idiomatic nature of the expression is preserved, as in *elle a un sacré coup dans l'aile* (lit. she has a hell of a hit in the wing). Other adjectives, e.g. *long, rouge* (lit. long, red) cannot be inserted. In G:*den (schönen) Schein wahren* ('to keep up appearances', lit. the (nice) pretence preserve) the presence or absence of the adjective does not change the meaning at all, whereas any modification of G:*das Handtuch werfen* ('to throw in the towel', lit. the towel throw) would lead to a loss of the idiomatic meaning, and the expression would revert to a literal phrase.

Structural Flexibility For instance, *to keep tabs on* can be passivized without loosing its idiomatic meaning. While most verbal MWLs in French do not allow topicalization, F:*chercher midi à quatorze heures* ('to nit pick', lit. search midday at fourteen o'clock) can be topicalized: *Midi, il ne le cherchait pas à quatorze heures.* Similarly in German, topicalization of lexically fixed components is only rarely possible whereas standard word order variation applies to all verbal MWLs. G:*Jan hat dabei den Vogel abgeschossen* ('Jan surpassed everyone in this', lit. Jan has with it the bird shot) can be topicalized as in G:*den Vogel dabei hat dann Jan abgeschossen* ('finally, Jan surpassed everyone in this', lit. the bird with it has then Jan shot).

As said above, a formal description of MWLs should license all the allowed variations but exclude all non–allowed variations. In the next section, we show how to achieve this by writing local grammar rules for MWLs in the form of IDAREX REs.

3 Encoding MWLs with IDAREX

The IDAREX formalism (**ID**ioms **A**s **R**egular **EX**pressions) which uses a two-level morphology has been developed as part of the Finite State Compiler (FSC) at Rank Xerox Research Centre by L. Karttunen, P. Tapanainen and G. Valetto[1].

3.1 Morphological Variation

Because IDAREX uses a two–level morphology, words can be presented either in their base form at the lexical level or in an inflected form at the surface level. The *surface form* is preceded by a colon and restricts occurrences of the word to exactly this form, e.g.

```
:Welle
```

The *lexical form* is followed by an IDAREX morphological variable specifying morphological features of the word, and a colon. The *morphological variable* can be very general, such as `'A'` for any adjectival use, or more specific, such as `Abse` for adjectives that may not be used in comparative form and `Nsg` to restrict nouns to the singular. For example,

```
durchschlagend A:
```

[1]For a more detailed description of the formalism see Karttunen and Yampol (1993), Tapanainen (1994), Karttunen (1995), and Segond and Tapanainen (1995).

represents any occurrence of the word G:*durchschlagend* (lit. making its mark on sth) as adjective, e.g. G:*durchschlagende, durchschlagender, durchschlagendsten*, whereas in the following example, comparative and superlative form G:*grünere, grünste* are excluded for the adjective:

```
grün Abse:  Welle Nsg:
```

This way, the restricted morpho–syntactic flexibility of MWLs can be expressed in a compact way, relieving the lexicographer from the burden of listing in an exhaustive way all the possible forms in which an MWL may be used.

3.2 Modification

If an MWL can be modified with a particular word this is represented as an optional expression in parentheses, as in

```
:den (:schönen) :Schein
```

The definition of *word–class variables* allows to express lexically unrestricted modifications of an MWL such as insertion of any adverb(s) (the Kleene star operator indicates that the item may occur any number of times):

```
perdre V: ADV* :la :tête
```

On the basis of simpler word–class variables more complex ones may be defined for complex syntactic categories such as NP, ADVP or PP. The morphological and word-class variables can be seen as shortcuts for particular regular expressions which cover specific sequences of morphological and lexical information. Their definition depends on the exact form of the morphological analysis.

3.3 Lexical and Structural Variation

The formalism provides a set of *RE operators* to combine the descriptions of single words. Square brackets '[ ]' and the bar '|' are used to describe lexical variants and more complex alternations such as word order variation in German. For instance, for the French example above we write

```
perdre V: ADV* [:la :tête | :la :boule | :les :pédales ] ;
```

To express German verb–front and verb–final word order as in
V1/2: *dabei wahrt er (immer) den Schein* ('in this, he (always) keeps up appearances')
Vend: *um den Schein zu wahren* ('in order to keep up appearances')
we write

```
[ wahren Vfin: (ADV* NPnom) ADV* :den (:schönen) :Schein |
 :den (:schönen) :Schein (:zu) wahren V: ] ;
```

As such REs quickly become very long and complicated, IDAREX allows the definition of *macros* to capture generalisations on the syntactic level.

3.4 Macros

The power of macros resides mainly in their parametrization. Any position in the macro that we want to instantiate differently for each use is indicated by a parameter `$i`. Instantiations of parameters can be single words in lexical or surface form, variables, operators or other macros. A Macro can be inserted within any IDAREX expression by using its name and specifying,

between parentheses and separated by spaces, the parameters that must be substituted to the place-holders at the moment the macro is expanded by the IDAREX compiler.

With the macro mechanism, regularities that can be observed for a whole class of MWLs need to be described in detail only once. The defined macro can then be applied to any instance of this class, using a simple and short expression. This offers a lot of flexibility and the possibility to represent complex phenomena in a compact way. General syntactic properties of MWLs that have to be accessible already at the lexical level can also be expressed.

We want to illustrate the usefulness and expressive power of macros with standard word order variation of German verbal MWLs. Three basic orderings are possible, verb-first (V1), verb-second (V2) and verb-final (Vend), and an infinitive construction (Inf+zu). In addition, scrambling of the external argument (*dir/Ute*) with the subject (*Jan*) is possible in all three word orderings.

V1: *Kommt Jan dir dabei in die Quere?*
('does Jan cross your path in this?', lit. comes Jan you with it in the widthways)

V1Scr: *Kommt dir dabei Jan in die Quere?*
('does Jan cross your path in this?', lit. comes you with it Jan in the widthways)

V2: *Jan kam ihr dabei in die Quere.*
('Jan crossed her path', lit. Jan came her with it in the ...)

V2Scr: *Ihr kam dabei Jan in die Quere.*
('Jan crossed her path', lit. her came with it Jan in the ...)

Vend: *Ute war sauer, weil Jan ihr dabei in die Quere kam.*
('Ute was angry because Jan had crossed her path')

VendScr: *Ute war sauer, weil ihr Jan dabei in die Quere kam.*

Inf+zu: *Jan versucht, Ute dabei nicht in die Quere zu kommen.*
('Jan tries not to cross Ute's path')

All these variations of G:*in die Quere kommen* ('to cross sb's path', lit. in the widthways come) are covered with the following RE using the macro `WOV1ARG` with three parameters, one for the type of the external argument (which could be a dative or accusative object or a PP), the second one for the lexically fixed components of the MWL, and the third one for the verbal component:

```
WOV1ARG(NPdat fix3(:in :die :Quere) kommen) ;
```

The instantiation of the second parameter uses another macro, `fix3`. This is one of a group of auxiliary macros `fix_i` that we use because we want to be able to instantiate the second parameter of `WOV1ARG` with expressions of variable length:

fix5: `$1 $2 $3 $4 $5` fix3: `$1 $2 $3` etc.

`WOV1ARG` is defined as

```
[ $3 Vfin: 1ARG1($1) $2 | $2 (Vaux ( 1ARGEND($1) )) (:zu) $3 V: ]
```

using further macros `1ARG1` and `1ARGEND`. The first part of the disjunction covers V1 and V2 order, the second part matches verb–final order and the infinitival construction and in addition topicalization of the fixed MWL component (*in die Quere*). `1ARG1` and `1ARGEND` capture regular syntactic properties of German and can be reused in other macro definitions: they take care of

scrambling of subject and external argument for V1/V2 and Vend ordering, respectively (V1 vs. V1Scr, V2 vs. V2Scr, and Vend vs. VendScr in the examples above).

`1ARG1` is defined as

```
[ (ADV* NPnom) ADV* $1 | (ADV* $1) ADV* NPnom ] ADV*
```

`1ARGEND` is defined as

```
ADV* [ NPnom ADV* $1 | $1 ADV* NPnom ] ADV*
```

In both definitions, `$1` is instantiated with the variable for the subcategorized idiom–external complement, i.e. `NPakk, NPdat`, or with a macro for PPs[2]. In `1ARG1`, the first component in each part of the disjunction is made optional because in V2 order, this component is moved to topic position to the left of the verb.

The macro `WOV1ARG` covers word order variation of verbal MWLs with one external argument. For verbal MWLs with two external arguments (e.g. G:*jdm etw in Rechnung stellen* ('to charge sb for sth', lit. sb sth in bill put)), another macro has to be used. The above example for the use of `WOV1ARG` leads to the following RE when the macro is compiled:

```
WOV1ARG(NPdat fix3(:in :die :Quere) kommen);
```

expands to

```
[ kommen Vfin: [(ADV* NPnom) ADV* NPdat | (ADV* NPdat) ADV* NPnom] ADV*
:in :die :Quere | :in :die :Quere (Vaux ([ NPnom ADV* NPdat | NPdat ADV*
NPnom ] ADV*)) (:zu) kommen V: ] ;
```

For German, further macros are defined for verbal MWLs where the fixed component can be scrambled with an idiom-external PP argument, for MWLs with a reflexive verb, for prefix verbs in their separated form, and for MWLs with a separable prefix verb. In French, macros describe for example the verb complex for reflexive verbs and constructions with the pronoun *en*.

4 Discussion

4.1 Comparison With Other Approaches

NLP treatments of MWLs in so–called high level grammar formalisms have for example been proposed in Abeillé and Schabes (1989) in the framework of lexicalised TAGs, in Erbach (1992) and Copestake (1994) for HPSG, in Van der Linden (1993) for CG. These approaches to our knowledge cannot satisfactorily represent lexical variants, nor the restricted flexibility and modifiability of MWLs. As MWLs are not simply lexical patterns but also show syntactic behaviour, their existence pleads for a tighter interaction between syntax and lexicon than has often been assumed. We have illustrated how this can be achieved using a finite–state approach and local grammar rules formalized as regular expressions.

Instead of using a high–level grammar formalism we describe MWLs with finite–state local grammars. Although finite–state techniques are known to be unable to represent all the dependencies found in natural language, they have the advantage of allowing a very efficient treatment of a great number of phenomena and the implementation of robust, large–scale NLP systems. Within the finite state approach to NL parsing, local grammars are finite state networks that

[2]The `PP` macro can be used to express the generally possible alternation of a PP with a pronominal adverb, e.g. *mit etw* vs. *damit*.

are matched against (parts of) a NL string. Informally speaking, if the string corresponds to one traversal of the network, it is recognised by the local grammar represented by that network. However, the use of these techniques is usually hampered by the unwieldiness in notation that they lead to. The development of adequate notational devices and corresponding compilers is therefore essential.

The presented approach overcomes this problem: instead of having to specify local grammars directly as finite state networks or as graphs as in work done by the 'Parisian School' (e.g. Maurel 1993, Roche 1993 and Silberztein 1993), they are compiled from descriptions which take the form of REs. IDAREX REs provide a convenient way to mix inflected and uninflected word forms, morphological features and complete word classes, thus greatly relieving lexicographers from the burden of explicitly listing all the possible forms. With macros, generalizations about patterns that can occur for a whole class of MWLs can be expressed. This compactness and flexibility are, as far as we know, specific to our approach.

Encoding the local grammars as REs instead of encoding them directly as networks does of course not change the expressive power of the formalism, but it conveniently abstracts the handling of MWLs from the graph manipulation level, allowing to develop and employ devices that operate on string representations and map them to the underlying finite state networks. As we have shown above, this simplifies considerably the description of the different patterns of variation occuring in MWLs.

Another interesting property of the finite state approach in general is its modularity. This is an advantage our approach shares with the Parisian notion of local grammars. The local grammars we propose encode MWL restrictions to the general syntax, but they are not meant to replace the general syntax. For instance, in a local grammar rule such as

```
prendre V: (POSCLITIC) ADV * (NP) :sous POSS :bonnet ;
```

which stands for F: *prendre qch sous son bonnet* ('to make something one's concern', lit. take sth under one's hat) the inserted NP and adverbial constituents are external to the MWL and as such should be handled by the general syntax. If desired, the general syntax of an NP can again be a local grammar as for instance proposed in Silberztein (1993) where all linguistic phenomena are treated as local grammars; but it need not to be used in this way. In fact, contrary to the Parisian philosophy, we intend to investigate the integration of local FS grammars with other more powerful grammar formalisms.

4.2 Limitations of Local Grammars

Our matching approach differs from high level parsing where the goal is to constrain the grammar to match all and nothing but grammatical input. The local grammar rules we write are formulated as generally as possible, allowing for overgeneration. They cover at most sentence–length patterns. Although more specific and restrictive rules could be written, this is unnecessary as long as the input is grammatical. If the input were not grammatical the most absurd matches could be obtained. E.g. the pattern for G:*in die Quere kommen* defined in section 3.4 will also match the following ungrammatical strings of German:

* *Kamen oft dem Mann in die Quere.*
(lit. came-pl often the-DAT man in the widthways)

* *Das Kind kam aber dem Mann nicht das Wetter in die Quere.*
(lit. the child came but the-DAT man not the weather in the widthways)

* *In die Quere wollten ihm unter Umständen ich in aller Eile zu kommen.*
(lit. in the widthways wanted him under circumstances I in all hurry to come)

In fact, even grammatical input can sometimes be matched wrongly. To take a very simple example, in F:*cela vous coûtera 3 francs en plus* ('that will cost you 3 francs more') the pattern matched by the RE '`:en :plus`' is associated with the translation 'extra, more, too many'. In F:*la vie est de plus en plus chère* ('life is more and more expensive') the same RE matches part of the longer expression *de plus en plus* ('more and more') which is represented as '`:de :plus :en :plus`'. For this reason, applications employing our system to match their input against IDAREX representations of MWLs should always take care to select the longest among multiple matches. It is not sure, however, that the relation between two matching MWLs is always that of simple inclusion of the one in the other.

The technique has its limits: productive variations of MWLs cannot be systematically captured with our approach because such variations are by their very nature unforeseeable. Examples of this are ad–hoc compound formations, or combinations of metaphors and idioms, as in

> G:*das bißchen Kopf, das sie noch haben, zerbrechen sie sich mit ...*
> (from Fleischer 1982)
>
> ('with that little intelligence they still have they rack their brains over ...',
> lit. the little head that they still have break they themselves with ...)
>
> ← G:*sich den Kopf zerbrechen* ('to rack one's brains', lit. oneself the head break)
> + G:*Köpfchen haben/etwas im Kopf haben* ('to have brains', lit. little head have/sth in the head have)

While it is possible to extend the technique so that the system matches also the approximate *das bißchen Kopf haben* and *sich das bißchen Kopf zerbrechen* to the two expressions involved, it will continue to fail on sentences like the one used in last year's French presidential campaign where the candidate of the extreme right dismissed his more to the left competitors by saying F:*c'est bonnet rouge et rouge bonnet* (lit. it is bonnet red and red bonnet). This variation of the standard expression F:*c'est bonnet blanc et blanc bonnet* ('it's six of one and half a dozen of the other', lit. it is bonnet white and white bonnet) is prompted by the association of *red* with 'to the political left'. Of course, the exploitation of this type of information is not only beyond the scope of the approach that we have presented here but also beyond that of more powerful mechanisms.

5 Application: Context-Sensitive Dictionary Look–Up

The presented approach is successfully used in the COMPASS project[3] whose main aim is the development of an on–line foreign language comprehension tool. For this, standard bilingual dictionaries have been converted into special dictionary databases. The COMPASS system, based on the LOCOLEX engine developed at RXRC (see Bauer, Segond and Zaenen (1995) for a description), allows look–up of words in the bilingual dictionary database directly out of an on–line text. When the user clicks on an unknown word in the foreign language, LOCOLEX evaluates the sentence context of the queried word. Currently, the system determines the word's part of speech and whether the word is part of an MWL. In the latter case, the translation for the entire MWL is returned, otherwise a selection of translations for the correct part of speech. A more detailed description of the COMPASS system and the recognition of MWLs can be found in Breidt and Feldweg (1996).

[3] 'Adapting bilingual dictionaries for COMPrehension ASSistance', LRE-62-080.

In order for COMPASS to recognize MWLs these are represented as IDAREX REs in the dictionary database. Unfortunately, the dictionary conversion and adaptation cannot be done fully automatically, and in particular the formal representation of the many MWLs as REs is quite a time–consuming task. However, once the MWLs listed in the dictionary have been manually changed into their *canonical base form* — which includes possible lexical variants and modifiers and indicates morphologically flexible components and the scope of alternative components — the IDAREX REs describing all possible contexts in which the MWLs can occur can be produced automatically[4]. For instance, the following canonical forms correspond to some of the examples from section 2:

out of the blue	kick° the bucket
faire° le° fou°	durchschlagender° Erfolg°
grüne° Welle (sg)	perdre° ˆ la tête/la boule/les pédales ˆ
avoir° un (sacré) coup dans l'aile	den (schönen) Schein wahren°
das Handtuch werfen°	T: den Vogel (bei etw) abschießen°

Such canonical base forms, somewhat similar in spirit to the notation used in Longman's 'Dictionary of English Idioms' (1979), do not only form the basis for the automatic processing and recognition of MWLs. Human users as well would profit from a careful description of the variability of MWLs, so it should be worthwhile to also include the canonical forms in dictionaries for human users.

6 Conclusion

We believe the contribution of IDAREX to the problem of recognizing and handling MWLs in NLP systems is twofold. On the one hand, at the description level, the formalism allows to represent idiosyncratic properties as well as regularities of MWLs in a convenient, compact and general way. Its syntax is based on and exploits the well-known concept of REs, supplemented by extensions such as variables and macros. With this notation, it is possible to write local grammar rules that capture a great deal of different variations of MWLs. IDAREX can be easily employed by lexicographers, even with little specific training, and allows to represent large numbers of MWLs with a reasonable effort. So far, we have successfully applied this approach to more than 15,000 English, French and German MWLs (see also Segond and Breidt 1995). Moreover, the production of IDAREXREs can be supported by simple semi-automatic procedures, as we briefly described above.

On the other hand, at the operational level, IDAREX constitutes an efficient and modular mechanism for MWL recognition that can be employed in real life systems, due to the underlying two-level morphology and finite state technology. The transformation of regular expressions into finite state networks is completely disjoint from lexicographic work, and is carried out by the FSC compiler. Moreover, computation of matches between instances of MWLs found in texts and their finite state representations has in general a very good performance.

The successful use of IDAREX in a multi–lingual context as part of the COMPASS comprehension assistant has greatly contributed to the construction, optimization and validation of our paradigm and of the underlying finite state mechanism. The satisfactory results and the observations coming from such a fairly large scale field test are encouraging to further investigate and continue perfecting our approach to MWL recognition.

[4]The automatic RE production from canonical forms is so far only implemented for German, but the mechanisms can just as well be applied for the other languages

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